

# 1 A Mamdani fuzzy inference framework for graded pure-tone audiometric classification: development, validation and open-source implementation

## Abstract

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**Background and Objective.** Pure-tone audiometry is summarised by the pure-tone average and graded against fixed severity boundaries, which forces a binary verdict on thresholds that carry graded information. A Mamdani fuzzy inference system was developed to return a continuous index, and validated against the six-category Clark PTA-4 schema.

**Methods.** Three NHANES cycles administering air-conduction audiometry to adults aged 20 to 69 years were combined and split at the participant level. Six trapezoidal membership functions were constructed as a Ruspini partition, so memberships sum to one at every threshold, and a 42-rule Mamdani system produced a Fuzzy Audiometric Index. Agreement was assessed by weighted kappa, stratification by distance from a boundary, decision-curve analysis, and a probabilistic benchmark scored by Brier score, log loss and calibration error.

**Results.** Among 19,623 ears from 9,843 adults, the FAI achieved kappa 0.946 against the reference grade, 96.9% overall agreement and a mean absolute error of 4.32 dB, with 99.2% agreement for clear cases and 86.3% within 5 dB of a boundary. Macro-averaged sensitivity was 0.891 and macro-averaged specificity 0.992, against 96.9% overall agreement that the normal class, 89.1% of ears, largely determines. An ordinal logistic model recovered the reference grade exactly in all 3,913 test ears with a Brier score of zero, establishing that the reference label is a deterministic thresholding of its input.

**Conclusions.** The framework reproduces the reference classification while preserving the gradation that crisp thresholds discard. Because the reference label is a deterministic function of the pure-tone average, agreement cannot evidence the graded representation's value; a seven-frequency learner outperforms the index on accuracy and calibration, locating its contribution in interpretability and a graded output rather than in loss minimisation.

**Keywords:** audiometry, hearing loss classification, fuzzy inference, Ruspini partition, reproducibility, clinical decision support

## 2 Introduction

Pure-tone audiometry remains the primary quantitative assessment of hearing, reporting thresholds at octave and inter-octave frequencies from 0.25 to 8 kHz. These seven or eight values are condensed into a single pure-tone average and graded against fixed boundaries, most commonly the six-category Clark scheme [1] or the five-category WHO tables [2]. Such conventions are useful because they are simple and embedded in reporting standards, disability assessment and hearing-aid candidacy criteria worldwide.

Its limitations are well established. A continuous quantity is forced into six discrete bins, so one decibel can change a diagnosis while a meaningful difference within a bin is invisible. The boundaries are administrative, not physiological: the  $\pm 5$  dB test-retest

variability of audiometry exceeds the distance between adjacent categories, so many ears sit within measurement noise of a verdict. The PTA also discards configuration, so a flat loss and a notched one can share an average while implying different pathologies, and a label cannot express trajectory.

Approaches that soften these boundaries have been proposed. Machine-learning systems that digitise audiogram images classify the same crisp severity levels without an inspectable decision [3]. Data-driven profiling recovers phenotypes without an interpretable decision [4,5]. These methods answer a different question: the difficulty is not predicting the label more accurately but representing the quantity so it preserves gradation and stays legible to a clinician.

Fuzzy set theory fits that requirement: membership functions express partial belonging rather than binary assignment, and a Mamdani system [6] states its rules in terms a clinician can read and modify. Fuzzy approaches have been applied across clinical diagnosis [7] and to EEG-based dementia staging [8], but no published system integrates severity classification, configuration typing, asymmetry and temporal tracking in one inspectable framework validated on population-level data.

This study develops such a framework and validates it on a large nationally representative cohort. The objectives were to construct severity memberships forming a formal Ruspini partition; to validate the index on a held-out participant-level split; to benchmark its graded output against probabilistic models using proper scoring rules; to determine how the width of the graded region affects agreement; and to test how far the results depend on the classification schema rather than the method.

## **3 Materials and Methods**

### **3.1 Study Design and Data Source**

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This was a cross-sectional secondary analysis of audiometric data from three National Health and Nutrition Examination Survey (NHANES) cycles that administered air-conduction pure-tone audiometry to adults aged 20-69 years: 1999-2000 (AUX1), 2011-2012 (AUX\_G) and 2015-2016 (AUX\_I). Thresholds were recorded at 500, 1000, 2000, 3000, 4000, 6000 and 8000 Hz in each ear.

### **3.2 Study Sample and Data Cleaning**

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Ears with NHANES sentinel codes (666 no response at maximum output, 777 refused, 888 could not obtain, 999 did not come) and missing-value placeholders were excluded, and the valid range was clipped to  $-10$  to  $120$  dB HL. Ears with complete thresholds at all seven frequencies formed the analysis set.

### **3.3 Train-Test Split**

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The clean-ear dataset was split at the participant level into training (80%) and held-out test (20%) sets, so both ears of a participant remained in the same partition and bilateral correlation could not leak across the split; the split used a fixed seed and no stratification. Membership functions, optimization, rule development and FAI-to-label

calibration were performed on the training set alone, and all reported metrics on the held-out test set.

### 3.4 Membership Function Construction

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Thresholds were assigned to six severity categories and a trapezoidal membership function [a, b, c, d] was fitted per category on percentile criteria, with cores at  $b = P25$  and  $c = P75$  across all training-set thresholds. The categories follow the Clark scheme at 25, 40, 55, 70 and 90 dB HL [1], the convention in most clinical reporting; Clark defined those tiers on the three-frequency average, and they are applied here to the four-frequency window that WHO later adopted for grading. Algorithm 1 states the construction formally: cores are held at their empirical percentiles, adjacent shoulders are set to meet, and any interior gap below the floor is widened symmetrically about its midpoint. The floor is thus satisfied without moving a core, which is what preserves the sum at unity. WHO’s own grade tables use five categories at 20 dB steps [2,9] and its 2021 revision uses seven at 15 dB steps [10]; the 2021 schema was validated alongside and is reported as a companion analysis.

The six trapezoids were arranged as a formal Ruspini partition [11], so their memberships sum to exactly 1 at every point of the 0 to 120 dB universe and no threshold is unclassified. Retaining the percentile cores, each trapezoid’s left foot was set to the previous category’s core end and its right foot to the next category’s core start, making adjacent shoulders complementary linear ramps that cross at 0.5 at the midpoint of the core gap.

A partition’s transition width equals its core gap, which a data-driven fit can leave narrower than the  $\pm 5$  dB test-retest variability the graded region represents. The construction therefore accepts a minimum transition width, set to 10 dB by widening the gaps while preserving the partition property; this floor was evaluated by ablation.

### 3.5 Configuration and Asymmetry Fuzzification

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Audiogram slope, the decibel change from 500 Hz to 4 kHz, was fuzzified into five categories: Rising ( $< -5$  dB), Flat ( $-5$  to 12 dB), Gently Sloping (12–28 dB), Steeply Sloping (28–45 dB) and Precipitous ( $> 45$  dB). A notch module quantifies the 4 kHz dip relative to the mean of the 2 kHz and 8 kHz thresholds, fuzzified into No Notch, Shallow Notch ( $< 15$  dB) and Deep Notch ( $\geq 15$  dB).

Asymmetry was fuzzified into Symmetric ( $\leq 15$  dB), Mildly (16–30 dB), Moderately (31–45 dB) and Severely Asymmetric ( $> 45$  dB). The operational criterion is the maximum absolute difference across frequencies, more sensitive than either definition Suen et al. evaluated: the AAO-HNS criterion of a pure-tone-average difference above 15 dB, and the Veterans Affairs criterion of 20 dB or more across two contiguous frequencies or 10 dB or more across three [12]; no standard criterion for asymmetry exists.

### 3.6 Fuzzy Inference System

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A Mamdani-type fuzzy inference system [6] was constructed with 42 expert-derived rules in three functional groups: 12 severity rules mapping fuzzified thresholds to the severity output, 14 configuration rules classifying audiogram shape, and 4 complex-interaction

rules for contour presentations the severity group does not capture. A further 12 rules map inter-aural asymmetry to a separate referral output rather than to severity, since asymmetry indicates unilateral pathology rather than a poorer-hearing ear. The 42-rule base serves both single-ear and bilateral input, so the validated configuration and the deployed one coincide; under the single-ear protocol the asymmetry input is zero and the referral output is inert.

Inference used max-min composition, fuzzy union aggregation and centroid defuzzification to obtain a continuous Fuzzy Audiometric Index (FAI) on a 0 to 100 scale. The severity rules take the PTA-4 speech-frequency average as their primary threshold input, matching the reference standard; this keeps the FAI comparable to the standard against which it is validated, but means agreement with PTA-4 is partly by construction. Frequency-resolved information enters through the slope, notch and asymmetry inputs and through the per-frequency membership vectors.

### 3.7 Probabilistic Comparator Models

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Because a crisp label carries no graded information, the fuzzy output was benchmarked against probabilistic multi-class models scored by proper scoring rules rather than agreement alone. The FAI's graded output is its six Ruspini memberships evaluated at the primary threshold. Those values are non-negative and sum to one, so they constitute a probability distribution over severity and can be scored as one.

Two feature sets were used. The primary threshold alone matches the information the fuzzy severity rules consume, while the seven frequency thresholds provide the full audiogram and contain the four thresholds defining the reference label. Six models were fitted on training data and evaluated on the test set: proportional-odds ordinal logistic regression [13] and multinomial logistic regression on each feature set, plus gradient boosting and a random forest on the seven-frequency set, scored by multi-class Brier score, log loss, expected calibration error and quadratic-weighted kappa. A uniform distribution provided a reference floor.

### 3.8 Comparator Schemas

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The categories are a labelling convention rather than part of the method, so the pipeline was rebuilt under the WHO 2021 schema (seven categories at 20, 35, 50, 65, 80 and 95 dB HL) using the same cohort, split, core fitting, construction and calibration. The two arms differ only in schema. The six-category system was additionally scored against the 2021 reference, quantifying the cost of a mismatch between fitted categories and the boundaries claimed.

### 3.9 Evaluation Metrics

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Primary metrics were weighted Cohen's  $\kappa$  (quadratic weights), mean absolute error, and agreement stratified by borderline ( $\pm 5$  dB of a boundary) versus clear-case ( $\geq 6$  dB) status. Bland-Altman analysis compared FAI with PTA-4 after a linear transfer fitted on the training partition mapped the index onto the decibel scale. Per-class sensitivity, specificity and precision were reported to expose the contribution of the majority class to the aggregate figures.

Decision-curve analysis [14] compared FAI-based with PTA-4-based referral across threshold probabilities from 0.05 to 0.30, with the risk model fitted on the training partition. For each threshold probability  $t$ , net benefit was  $TP/n - FP/n \times t/(1-t)$ , with treat-all and treat-none providing reference curves, and two outcomes were evaluated.

### 3.10 Temporal Tracking

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The NHANES cycles analysed here are repeated cross-sectional surveys with no repeat audiometry on any participant, so no longitudinal audiometric sub-cohort could be constructed and the temporal tracking module, which applies fuzzy-adapted ASHA ototoxicity criteria [15] to FAI trajectories, is demonstrated on illustrative scenarios (Figure S1). Prospective validation would require serial audiometry in a monitored population, including extended high-frequency testing from 9 to 20 kHz, where ototoxic change appears first.

### 3.11 Statistical Analysis and Software

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Analyses used Python: scikit-fuzzy v0.5.0 for inference, scikit-learn v1.9 and statsmodels v0.14 for the comparators, SciPy v1.18 for testing. The complete analysis, including the membership construction, pipeline and verification scripts, is provided as executable open-source code.

### 3.12 Ethics Statement

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No ethics approval was required for this secondary analysis of de-identified, publicly available data.

## 4 Results

### 4.1 Cohort Characteristics

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The cohort comprised 10,889 adults aged 20-69. After exclusions, 19,623 ears from 9,843 adults had complete air-conduction thresholds at all seven frequencies: 15,710 ears (7,881 participants) in the training set and 3,913 (1,962 participants) in the test set. Thresholds were right-skewed with a floor effect at 0 to 10 dB HL, rising from 11.2 dB HL at 500 Hz to 27.4 dB HL at 8 kHz, consistent with age-related and noise-induced high-frequency loss. Under the WHO PTA-4 classification [2], 88.1% of ears were normal, 8.7% mild, 2.2% moderate, 0.6% moderately severe, 0.3% severe and 0.1% profound,

Nearly one in five ears (18.7%) fell within  $\pm 5$  dB of a boundary, where crisp classification is least certain.

### 4.2 Membership Function Construction

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The percentile-based optimization produced the Ruspini-partition functions shown in Figure 1 (parameters in Table S1), with transition bands of 13.6, 10.0, 10.0, 10.0 and 10.9 dB from normal to profound. Verification confirmed the memberships sum to 1 at every sampled threshold across the 0 to 120 dB universe, with a maximum deviation of zero. Adjacent memberships cross at 0.5 at 23.2, 40.4, 57.1, 71.1 and 90.5 dB (Figure 1);

the calibrated FAI-to-label cut points, fitted rather than geometric, sit at 23.8, 48.9, 63.8, 66.4 and 95.0.

### 4.3 Classification Agreement with the Reference Grade

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On the test set the FAI achieved weighted Cohen’s  $\kappa = 0.946$  against the Clark PTA-4 grade, with 96.9% overall agreement and a mean absolute error of 4.32 dB. Agreement 99.2% for clear cases ( $\geq 6$  dB from a boundary) and 86.3% within the borderline zone ( $\pm 5$  dB, 17.8% of test ears). Spearman’s correlation was 0.812, restricted in range because 89.1% of test ears are normal and the upper tail is compressed; kappa and MAE are therefore the more direct agreement measures.

Per-class performance is given in Table 1. Sensitivity was 0.978 for normal and 0.977 for mild, fell to 0.507 for moderate and recovered to 0.885 for moderately severe; macro-averaged sensitivity was 0.891 against the 96.9% overall figure, which the normal majority determines. Of 122 disagreements, 101 moved upward by one grade and 21 downward, a predominantly upward error structure that increases false-positive referrals.

### 4.4 Agreement and Distance from the Severity Boundary

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Agreement recovered with distance from the boundary (Figure 2b): 86.3% within  $\pm 5$  dB, 91.0% at 10 dB and 94.3% at 15 dB. Bland-Altman analysis (Figure 2a) compared the FAI with PTA-4 after a linear transfer fitted on the training partition mapped the index onto decibels,  $\text{FAI} = 0.890 \times \text{PTA} + 3.11$  ( $r = 0.887$ ). The sub-unity slope follows from centroid defuzzification, which returns the centre of mass of the aggregated output and compresses the ends of the universe: mean difference  $-0.13$  dB with 95% limits of agreement  $-11.5$  to  $+11.2$  dB, with a proportionate bias of 0.13 dB per decibel.

### 4.5 Probabilistic Comparison with Machine-Learning Models

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Table 2 reports the probabilistic benchmark. A proportional-odds ordinal logistic regression on the primary threshold alone reproduced the reference grade exactly in all 3,913 test ears, with a Brier score and log loss of 0.0000, as did the same model on all seven frequencies. The reference label is therefore not a prediction target but a deterministic thresholding of the pure-tone average, so agreement metrics cannot discriminate between models: any model able to represent monotone thresholds recovers it exactly, so residuals measure how well each approximation fits a step function.

Among the models that do not saturate, the FAI membership vector returned a Brier score of 0.0482, log loss 0.0810 and expected calibration error 0.0266, against 0.0136, 0.0302 and 0.0021 for gradient boosting on the richer seven-frequency input. The ensemble was better on every metric, so the index is not competitive with it on accuracy: read on one threshold rather than seven, its arg-max recovered the reference grade at kappa 0.9453.

## 4.6 Decision-Curve Analysis

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FAI-based referral maintained net benefit comparable to PTA-4-based referral across the threshold range for both outcomes, and both exceeded treat-all at every threshold (Figure 3); for any hearing loss at a 10% threshold, net benefit was 0.126 against 0.131 and 0.034 for treat-all. The small differences favour the crisp markers, which are defined on the outcome itself; the graded index does not sacrifice decision utility.

## 4.7 Transition-Width Ablation

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Flooring the core gaps at 10 dB against the raw percentile gaps, which ranged from 5.7 to 13.6 dB, moved weighted kappa from 0.9437 to 0.9460 (+0.0022, 95% CI -0.0014 to +0.0058), borderline agreement from 85.9% to 86.3% and clear-case agreement from 99.0% to 99.2%. The raw gaps were therefore already behaving like a floor near 4 dB, and the deployed setting buys a small, non-significant gain over them.

Sweeping the floor from 4 to 16 dB in 2 dB steps made the deployed setting the best-performing floor (Table 3): every setting except 4 dB differed significantly from 10 dB. Absolute error nevertheless fell monotonically with width, from 4.34 to 4.17 dB, so categorical and continuous agreement diverge: wider shoulders smooth the graded output while blurring the boundary.

## 4.8 Validation Under the WHO 2021 Schema

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Under the 2021 schema the framework performed better on every metric except raw overall agreement (Table 4): kappa 0.961 against 0.946, mean absolute error 3.69 dB against 4.32, and rho 0.840 against 0.800. Borderline agreement was 90.5%, measured over the harder population, since the 2021 boundaries place 29.7% of ears within  $\pm 5$  dB of one against 17.8%. Scoring the six-category system against the 2021 reference returned kappa 0.864.

## 4.9 Clinical Case Studies

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Four synthetic archetypes illustrate the framework (Figure 4). Case B is the most informative: a noise-induced notch with a normal PTA-4 of 22.5 dB returned FAI 21.2, while its configuration output identified a precipitous notched profile the severity grade alone misses.

# 5 Discussion

## 5.1 Principal Findings

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A Mamdani fuzzy inference system was developed and validated that maps pure-tone thresholds onto a continuous index while preserving the information crisp classification discards. On held-out NHANES ears the FAI achieved weighted kappa 0.946 and matched the crisp grade in 99.1% of clear cases. Three findings stand out: the system remains compatible with standard practice; the graded representation is a better classifier than the calibrated label built on it, its arg-max reaching 0.9564; and the wider graded region

improves rather than costs agreement, raising kappa by 0.019 and borderline agreement by 5.6 points with bootstrap intervals excluding zero.

## 5.2 Agreement Cannot Evidence the Graded Representation

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The probabilistic benchmark produced the key result. A proportional-odds ordinal logistic regression recovered the reference grade exactly in all 3,913 test ears with a Brier score of zero, which is not a strong model but proof the label is a deterministic monotone function of the pure-tone average. The mechanism is complete quasi-separation: the maximum-likelihood estimate diverges, having no interior optimum, so the fitted probabilities are exactly 0 and 1. Agreement metrics therefore saturate, so a graded representation cannot be shown to be better by kappa.

What survives is discrimination among approximations. Gradient boosting, on seven-frequency input, scored kappa 0.9832 against the memberships' 0.9453, with a lower Brier score (0.0136 against 0.0482), lower log loss (0.0302 against 0.0810) and better calibration (expected calibration error 0.0021 against 0.0266). Read on one threshold rather than seven the index is at an information disadvantage no construction removes. The probabilistic properties discriminate. What the index offers instead is an inspectable rule base, a graded output no crisp classifier provides, and explicit behaviour at the boundaries where graders disagree. Membership degrees are not probabilities: they express graded belonging to a category, the vagueness of a boundary, rather than the aleatory uncertainty of an outcome, and are scored as a distribution over grades rather than a belief about one.

## 5.3 The Classification Schema Is a Parameter

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The construction is not tied to the 25/40/55/70/90 boundaries, which trace to Clark. The same cores, partition and calibration applied to the 2021 boundaries produced a better-calibrated system with lower absolute error, so the framework is schema-agnostic and agreement figures are interpretable only against the schema they were computed with. A change to WHO's grade table would require recalibration, not redesign. A Takagi-Sugeno-Kang system would avoid the boundary compression of centroid defuzzification and map onto decibels directly, but it replaces the linguistic consequent with a numeric one, discarding the rule trace a clinician can read and a regulator can review; that interpretability is why a Mamdani architecture was retained.

## 5.4 Clinical Implications

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In triage the FAI could flag patients who straddle a severity boundary, so a borderline case is monitored rather than forced into a category that misstates it. Prescriptive methods such as NAL-NL2 [16] and DSL [17] already derive frequency-specific gain from the full audiogram, so the index is not a fitting prescription: its contribution is automated triage and remote screening, and a configuration summary for notched or sloping patterns the PTA obscures. For ototoxicity monitoring, the temporal module flags sub-threshold FAI changes without a PTA boundary crossing, an early warning conventional staging misses and that requires prospective validation.

## 5.5 Comparison with Prior Work

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The results are consistent with the computational audiology literature. Object-detection systems that read audiograms and label severity [3] achieve strong accuracy but are not inspectable, and clustering-based profiling [4,5] recovers phenotypes without an explicit decision a clinician can interrogate. This study adds that such comparisons are confounded when the target is a deterministic function of its inputs, and that graded outputs need proper scoring rules. No previously published fuzzy system integrates severity classification, configuration typing, asymmetry and temporal tracking in one inspectable framework validated with a participant-level split.

## 5.6 Limitations

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Only air-conduction thresholds were used, so the framework grades combined loss and cannot separate sensorineural from conductive or mixed components; the complex-interaction rules therefore capture contour rather than pathophysiological mixed loss. The rule base has not undergone formal Delphi consensus. The three cycles covered adults aged 20 to 69 years, so children and older adults are not represented and extended high-frequency audiometry was not covered.

Inter-aural asymmetry was decoupled from severity and now drives a referral recommendation instead. An ear with a 55 dB pure-tone average could previously be graded moderately severe because the contralateral ear differed; asymmetry is a referral indicator, not an amplifier of peripheral loss. The change is neutral for the figures reported here, since the single-ear protocol pins the asymmetry input to zero.

The World Report on Hearing [10] lowered the normal boundary to 20 dB HL and added a complete-loss category. The reference standard used here, and the frequency band the severity rules share with it, mean agreement is partly by construction. The benchmark quantifies that dependence rather than removing it, since the exact reconstruction shows the ceiling is definitional. This limits what any agreement figure can mean, and is why the graded outputs and probabilistic metrics carry the argument. The case studies are synthetic, not patient records. Whether the FAI predicts hearing-aid outcomes or surgical candidacy has not been assessed.

## 5.7 Generalizability and Future Directions

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Additional inputs can be accommodated by adding membership functions and rules, and a calculator with a REST API for EHR integration has been deployed for evaluation. Future work will validate the system prospectively, explore neuro-fuzzy approaches such as ANFIS [18] that learn rule parameters while remaining interpretable, and test whether the graded index improves on the crisp grade in predicting speech-in-noise performance or hearing-aid outcomes.

## 5.8 Conclusions

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The Fuzzy Audiometric Index provides a continuous, frequency-resolved, interpretable classification of hearing loss that preserves the gradation lost to crisp thresholds. Validated on held-out NHANES ears, it agrees substantially with the reference

classification while preserving borderline, configuration and graded information. The probabilistic benchmark shows the reference label is a deterministic thresholding of its input, so agreement cannot evidence a graded representation; a seven-frequency learner outperforms the index on accuracy and calibration, locating its value in interpretability and the graded output rather than accuracy.

## Highlights

Mamdani fuzzy inference provides graded classification of pure-tone audiograms.

Ruspini partition ensures memberships sum to unity across 0 to 120 dB HL.

Validated on 19,623 NHANES adult ears with participant-level splitting.

Reference label shown to be a deterministic thresholding of its input.

Open-source implementation with a REST API for EHR integration.

## CRedit Author Statement

Sanyaolu Ameye: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization.

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## Declaration of Generative AI Use

During the preparation of this work the author used a large language model assistant to support drafting, language editing and code documentation. No AI or machine-learning tool generated the underlying data, selected the cohort, or computed any reported statistic; the fuzzy inference system is a transparent, rule-based model with no learned parameters. The author reviewed and verified all content, including checking every reference against CrossRef or PubMed, and takes full responsibility for the work.

## Declaration of Competing Interest

The author reports no competing interests to declare. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

## Data Availability

The NHANES data analysed here are publicly available from the U.S. Centers for Disease Control and Prevention. The fuzzy inference system, the Ruspini membership function construction, the analysis pipeline and the verification scripts are available as an open-source implementation at <https://github.com/ameye/fuzzy-audiogram>.

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## Tables

Table 1. Classification performance on the held-out test set (n = 3,913 ears).

| Severity category | n     | Sensitivity (95% CI) | Specificity (95% CI) | Precision (95% CI)  |
|-------------------|-------|----------------------|----------------------|---------------------|
| Normal            | 3,486 | 0.978 (0.973-0.983)  | 1.000 (0.991-1.000)  | 1.000 (0.999-1.000) |
| Mild              | 310   | 0.977 (0.954-0.989)  | 0.974 (0.968-0.979)  | 0.763 (0.719-0.802) |
| Moderate          | 73    | 0.507 (0.395-0.618)  | 0.997 (0.995-0.999)  | 0.787 (0.651-0.880) |
| Moderately severe | 26    | 0.885 (0.710-0.960)  | 0.995 (0.993-0.997)  | 0.561 (0.410-0.701) |
| Severe            | 12    | 1.000 (0.757-1.000)  | 1.000 (0.999-1.000)  | 1.000 (0.757-1.000) |
| Profound          | 6     | 1.000 (0.610-1.000)  | 1.000 (0.999-1.000)  | 1.000 (0.610-1.000) |

Note. Wilson score intervals. Macro-averaged sensitivity 0.891 and macro-averaged F1 0.858 against 96.9% overall agreement, which the normal majority largely determines. Of 122 disagreements, 101 moved upward by one grade and 21 downward.

Table 2. Probabilistic benchmark on the held-out test set (n = 3,913 ears).

| Model                 | Features      | Weighted $\kappa$ | Accuracy | Brier  | Log loss | ECE    |
|-----------------------|---------------|-------------------|----------|--------|----------|--------|
| Ordinal logistic      | PTA-4         | 1.0000            | 100.0%   | 0.0000 | 0.0000   | 0.0000 |
| Ordinal logistic      | 7 frequencies | 0.9995            | 100.0%   | 0.0005 | 0.0071   | 0.0003 |
| Multinomial logistic  | PTA-4         | 0.9958            | 99.8%    | 0.0138 | 0.0297   | 0.0228 |
| Multinomial logistic  | 7 frequencies | 0.9953            | 99.7%    | 0.0097 | 0.0212   | 0.0142 |
| Gradient boosting     | 7 frequencies | 0.9832            | 99.1%    | 0.0136 | 0.0302   | 0.0021 |
| Random forest         | 7 frequencies | 0.9685            | 98.4%    | 0.0294 | 0.0540   | 0.0238 |
| FAI membership vector | PTA-4         | 0.9453            | 96.9%    | 0.0482 | 0.0810   | 0.0266 |
| Uniform reference     | —             | 0.0000            | 89.1%    | 0.8333 | 1.7918   | 0.7242 |

Note. The FAI row scores the six Ruspini memberships evaluated at the primary threshold as a probability distribution. ECE, expected calibration error. The baseline assigns every ear to the majority class, which is why its accuracy is 89.1% while its Brier score of 0.8333 and log loss of 1.7918 are those of a uniform distribution over six classes; a uniform classifier would score 16.7% accuracy. The ordinal logistic’s exact reconstruction shows that the reference label is a deterministic thresholding of the pure-tone average, so agreement metrics cannot discriminate between models on this task. Among models that do not saturate, the seven-frequency learners outperform the FAI memberships on every metric, so the index’s contribution is interpretability and a graded output rather than accuracy.

Table 3. Transition-width sweep. Same construction and same calibration at every setting; only the floor on the core gap differs. A 4 dB floor is equivalent to the raw percentile gaps, which ranged from 5.7 to 13.6 dB.

| Floor (dB)    | Weighted $\kappa$ | Borderline ( $\pm 5$ dB) | MAE (dB) | Difference vs 10 dB | 95% CI             |
|---------------|-------------------|--------------------------|----------|---------------------|--------------------|
| 4             | 0.9437            | 85.9%                    | 4.34     | -0.0022             | -0.0058 to +0.0014 |
| 6             | 0.9436            | 85.8%                    | 4.34     | -0.0023             | -0.0037 to -0.0010 |
| 8             | 0.9399            | 85.0%                    | 4.33     | -0.0060             | -0.0113 to -0.0005 |
| 10 (deployed) | 0.9460            | 86.3%                    | 4.32     | —                   | —                  |
| 12            | 0.9371            | 84.3%                    | 4.30     | -0.0089             | -0.0135 to -0.0042 |
| 14            | 0.9402            | 84.3%                    | 4.27     | -0.0058             | -0.0091 to -0.0029 |
| 16            | 0.9371            | 83.9%                    | 4.17     | -0.0088             | -0.0129 to -0.0052 |

Note. Paired bootstrap, 2,000 resamples over 3,913 test ears. The deployed 10 dB floor is the best performing setting; every other floor except 4 dB differs significantly from it. Mean absolute error falls monotonically with width while categorical agreement turns over, so the two diverge at the wide end, wider shoulders smoothing the graded output while blurring the boundary.

Table 4. Validation under the WHO 2021 schema against the Clark six-category scheme. Same cohort, split, percentile-core fitting, Ruspini construction and calibration; only the classification schema differs.

| Metric                             | Clark six-category (25/40/55/70/90) | WHO 2021 (20/35/50/65/80/95) |
|------------------------------------|-------------------------------------|------------------------------|
| Weighted kappa vs own reference    | 0.946                               | 0.961                        |
| Overall agreement                  | 96.9%                               | 96.7%                        |
| Borderline agreement ( $\pm 5$ dB) | 86.3%                               | 90.5%                        |
| Borderline share of test ears      | 17.8%                               | 29.7%                        |
| Clear-case agreement               | 99.2%                               | 99.3%                        |
| Spearman rho vs PTA-4              | 0.800                               | 0.840                        |
| MAE vs PTA-4 (dB)                  | 4.32                                | 3.69                         |

Note. For reference, scoring the Clark six-category system against the WHO 2021 boundaries returns kappa 0.864, so the categories must be fitted to the boundaries being claimed. The 2021 arm faces the harder borderline population, with 29.7% of ears within 5 dB of a boundary against 17.8%.

## Figure Captions

Figure 1. Trapezoidal membership functions for the six hearing loss severity categories, arranged as a formal Ruspini partition. Successive functions share their feet with their neighbours' cores, so adjacent shoulders are complementary linear ramps that cross at 0.5 and memberships sum to 1 at every threshold across the 0 to 120 dB universe. Transition bands measure 13.6, 10.0, 10.0, 10.0 and 11.2 dB from normal to profound.

Figure 2. (a) Bland-Altman plot of the difference between FAI and PTA-4 against their mean on the held-out test set. The FAI is a defuzzified index on a 0 to 100 universe rather than a decibel quantity, so it is mapped onto the decibel scale before differencing, using  $\text{FAI} = 0.890 \times \text{PTA-4} + 3.11$  fitted on the training partition ( $r = 0.887$ ); bias  $-0.13$  dB, 95% limits of agreement  $-11.5$  to  $+11.2$  dB. (b) Classification agreement as a function of distance from the nearest Clark severity boundary. The FAI matches the crisp grade on 99.2% of clear cases ( $\geq 6$  dB from a boundary) and returns graded membership on 86.3% of borderline cases ( $\pm 5$  dB). Figure 3. Decision-curve analysis on the held-out test set: net benefit of FAI-based referral versus PTA-4-based referral, the crisp reference grade rule, and treat-all/treat-none strategies, for (a) any hearing loss ( $\geq$  mild) and (b) moderate-or-worse loss. Risk models were fitted on the training partition. Figure 4. Clinical case panels: four synthetic archetypes (A borderline mild, B noise notch, C presbycusis, D asymmetric) showing PTA versus FAI labels, membership bar plots and configuration vectors. Coloured background bands indicate WHO severity categories. Threshold arrays are given in Table S5.